Lecture 3 Part 1:

Arguments, The Scientific Method

Ilyena Hirskyj-Douglas

Disclaimer

- Much of this first part is somewhat philosophical in nature
- This introduces some subjectivity into the contents
- “Get 3 academics together and ask for their opinion on a topic and you’ll get 4 answers”
- Nevertheless, it’s important to get you thinking about these topics

Contents

- An overview of the pre-recorded videos *(and what to look out for)*
- Arguments *(philosophically)*
- Formal and informal fallacies
- A note about the skills needed to do research

The pre-recorded videos this week

These, however, are not opinionated / subjective

Instead, these are focused on the scientific method

- An “empirical method for acquiring knowledge”
- Important to know this as the remaining weeks we will be discussing experimental design and other parts of this process/method

Video - Neil deGrasse Tyson's Analogy for the Scientific Method

Two important lessons within it surrounding:

- The importance of defending what you consider to be true
- The emergence of a scientific truth

Video - Neil deGrasse Tyson's Analogy for the Scientific Method

The importance of defending what you consider to be true.

- As a scientist, you should defend what you consider to be true but be open and willing to be proved wrong

- Note: you should defend your argument with evidence, theory, and reason — you should not rely entirely on opinion and belief

Video - Neil deGrasse Tyson's Analogy for the Scientific Method

The emergence of a scientific truth

- A single result is never enough — it requires verification
- Only through repeated observation and experimentation does a scientific truth emerge

Video - The Richard Feynman Videos

Richard Feynman was a physicist but his videos on the scientific method and philosophy are very well regarded and contain important lessons

- The importance of being proved wrong by your experiments/data
- The importance of guessing
- The importance of having specific theories to test
- That you never prove a theory is completely right — you can only ever be sure when we're wrong

Video - The Richard Feynman Videos

Suppose you develop a theory and you design an experiment to test it, and two competitors design an experiment to test it as well

- If all three experiments give the same result
- And if this result proves the theory is wrong
- You cannot simply conclude the theory is correct and the experiments are wrong

“The first principle is that you must not fool yourself and you are the easiest person to fool”

Video - The Richard Feynman Videos

As a scientist you must always be willing to be proved wrong

- The verification of a theory, at best, means that you are only temporarily right — that is, the theory has not yet been proved wrong
- But tomorrow's experiment may prove you wrong and show the error
- So, we are never right — we can only know for sure when we are wrong

Many of the videos introduced "The Scientific Method"

Let’s talk about arguments

Argument (philosophically)

A statement or group of statements called premises intended to determine the degree of truth or acceptability of another statement called conclusion

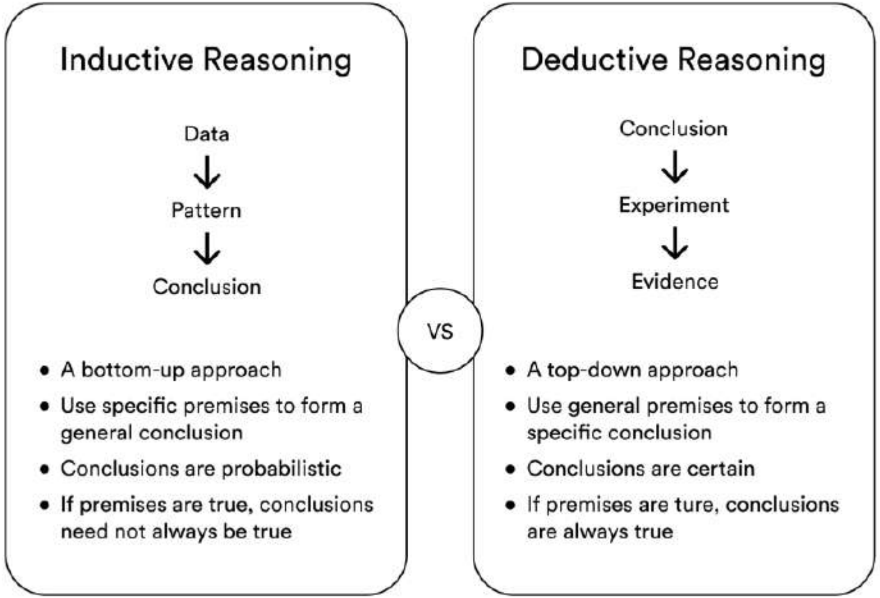
Scientific inference is the process of reaching conclusions based on available evidence and theories

- We have one (or more) premise, from which we draw a conclusion

Deductive inference vs Inductive inference

Deductive inference vs **Inductive inference**

- The main difference between **inductive** and **deductive** reasoning is that inductive reasoning aims at developing a theory, while deductive reasoning aims at testing an existing theory.



Argument (philosophically) — Deductive Inference

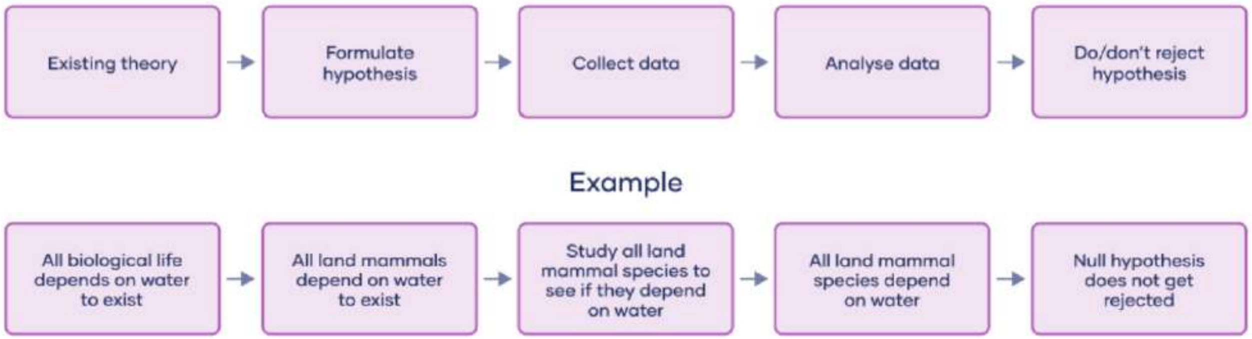
An inference is deductive if there is a relation between the premises and the conclusion guaranteeing the truth of the conclusion if the premises are true

- If the premises are true, the conclusion must be true

Deductive inference

Deductive reasoning is a logical approach where you progress from general ideas to specific conclusions.

It's often contrasted with inductive reasoning, where you start with specific observations and form general conclusions.



Examples: Deductive logic arguments

Premise	All insects have exactly six legs.
Premise	Spiders have eight legs.
Conclusion	Therefore, spiders are not insects.

Premise	Blue litmus paper turns red in the presence of acid.
Premise	The blue litmus paper turned red after I dropped some liquid on it.
Conclusion	Therefore, the liquid is acidic.

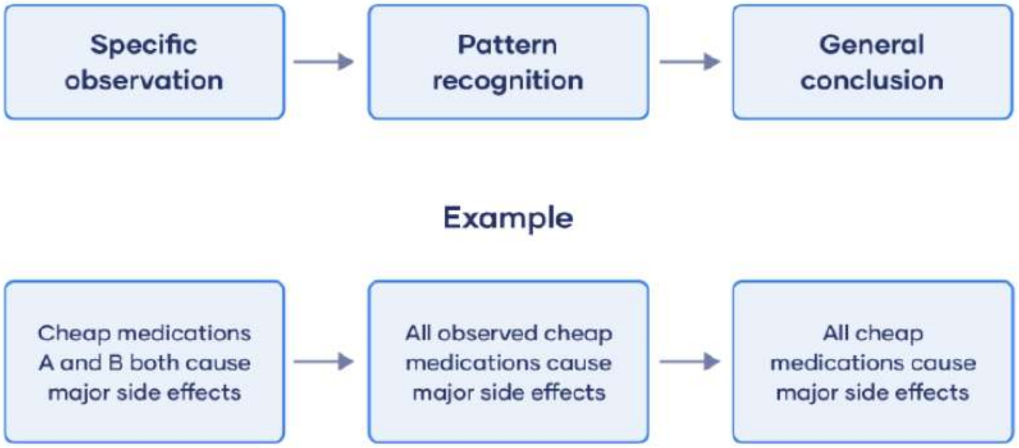
Argument (philosophically) — Inductive Inference

An inference is inductive if we use premises about some observed objects to draw conclusions about similar unobserved objects

- We observe a sample to draw conclusions about a population
- Inductive inference is what is mostly used in research

Inductive inference

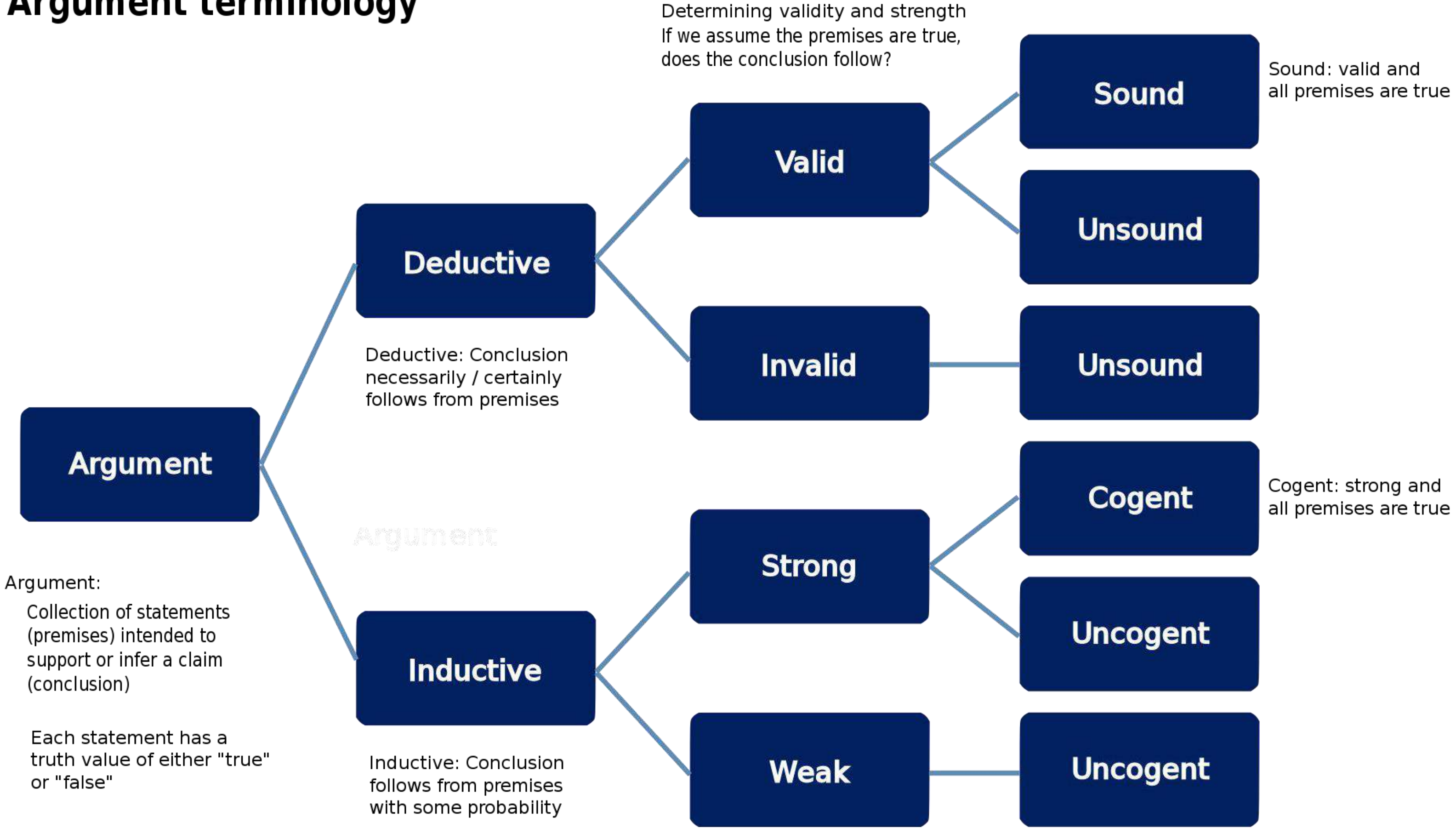
Start with one specific observation, add a general pattern, and end with a conclusion.



Examples: Inductive reasoning

Stage	Example 1	Example 2
Specific observation	Nala is an orange cat and she purrs loudly.	Baby Jack said his first word at the age of 12 months.
Pattern recognition	Every orange cat I've met purrs loudly.	All **observed** babies say their first word at the age of 12 months.
General conclusion	All orange cats purr loudly.	All babies say their first word at the age of 12 months.

Argument terminology



A fallacy in an argument

What is a fallacy?

A fallacy is the use of invalid or otherwise faulty reasoning in the construction of an argument (which may appear stronger than it really is if the fallacy is not spotted)

- Some fallacies are committed intentionally, others are accidental
- Fallacies are commonly divided into "formal" and "informal"

Formal fallacies

Formal fallacies

- A formal fallacy is a flaw in the structure of a deductive argument which renders the argument invalid
- Most formal fallacies are errors of logic
  - The conclusion doesn’t really is not supported by the premises
  - Either the premises are untrue, or the argument is invalid

Formal Fallacies: An Example

Premise: All black bears are omnivores
Premise: All racoons are omnivores
Conclusion: All racoons are black bears

- Both black bears and racoons are a subset of omnivore, but these subsets do not overlap — that fact makes the conclusion illogical

- This example is obvious, but this example can get dangerous quite quickly

Formal fallacies — dangerous arguments

Suppose someone made similar arguments over ethnicities and religions:

- Not everyone of the same ethnicity is the same religion
- Not everyone of the same religion is the same ethnicity

- This is not a potentially dangerous argument: one that could spread confusion, hate and misinformation.

Informal fallacies

Informal fallacies

An informal fallacy originates in an error in reasoning other than an improper logical form

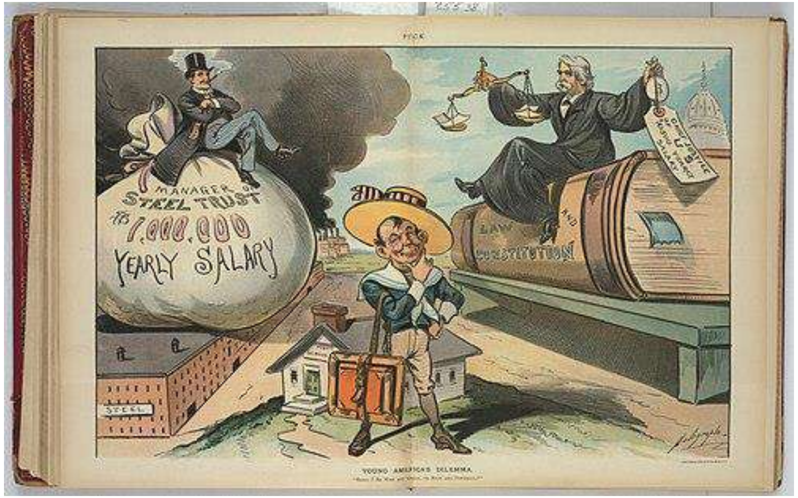
- Primarily due to the content and context of the argument
- Many types of informal fallacy — we'll discuss "false dilemmas" and "straw man arguments" in more detail

False dilemmas

False dilemmas

An informal fallacy based on a premise that erroneously limits what options are available

- Oversimplification of reality by excluding viable alternatives



Example here from Puck 1901: Shall I be wise and great, or rich and powerful?

Suggests only two choices when there can be many.. You could be poor and great for example

False dilemmas — Examples

- "Stacey spoke out against capitalism; therefore, she must be a communist"
  - One excluded option is Stacey may be neither communist or capitalist
  - The fact she spoke out against capitalism doesn’t make her a communist
    - she could be, but she could also be a lot of other things

False dilemmas — Examples

- "If we want to make campus safer, we can either install more security cameras or ban visitors"
    - One excluded option is to hire more on-site security, have them patrol around the campus
    - Maybe all the security staff are striking? To what extent are visitors the problem? Etc.

Straw man arguments

Straw Man Arguments

- A form of argument and an informal fallacy of having the impression of refuting an argument, whereas the real subject of the argument was not addressed or refuted, but instead replaced with a false one
  - One who engages in this fallacy is said to be "attacking a straw man"

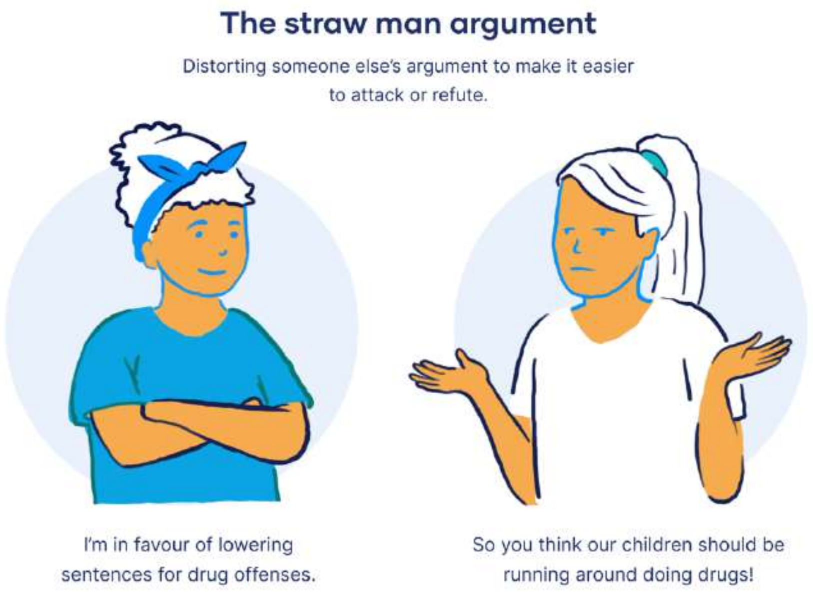
Straw Man Arguments

- The typical straw man argument creates the illusion of having refuted or defeated an opponent's proposition through the replacement of it with a different proposition
- And the subsequent refutation of that false argument instead of the opponent's proposition

Straw Man Arguments

In short:

- Person 1 asserts a proposition P
- Person 2 misrepresent this proposition P, then makes an argument against this misrepresented proposition



Straw Man Argument

How might proposition P be misrepresented?

- Quoting words out of context
  - For example, choosing quotations that misrepresent the opponent’s intentions
- Oversimplifying an opponent’s argument

Straw Man Argument

How might proposition P be misrepresented?

- Exaggerating an opponent’s argument
- Presenting someone who defends a position poorly as the defender, then denying that person's arguments
  - To give the appearance every upholder of this position (and thus the position itself) has been defeated

Straw Man Argument – Example One

Parent: “You aren’t allowed to go to the party tonight, you have an exam tomorrow”
Child: “Why do you hate me”

- The proposition says nothing about the parent’s love for their child
- But the child positions not going to the party as a measure of their parent’s love for them

Straw Man Argument – Example Two

Girlfriend: “I want to eat Chinese food instead of pizza tonight”
Boyfriend: “Oh so you hate pizza then?”

- The girlfriend says nothing about her opinion of pizza – just that her preference for this evening is to eat Chinese food
- A statement of preference alone doesn’t mean that the lesser preferred thing is disliked

Possibility vs Probability

Possibility vs Probability

Often in argument someone might be tempted to use the phrase “well is it possible” as an ultimate defense of their viewpoint

- Assumption being if something is possible then they cannot be wrong

This is not a good defense / position

- Often, the question is not “what is possible?” Instead, the question is “is it probable?”

Possibility vs Probability

- With enough creativity most things are possible
    - “If we assume A, B, C, and if X then Y then Z”
- But this does not mean it is probable

Possibility vs Probability

- Is it possible I will become the first person to establish a base on the moon?
  - If I quit my job lecturing and form a startup
  - Develop a successful product and become a billionaire
  - Befriend Elon Musk and invest heavily in SpaceX
  - Convince Elon Musk to establish a base on the moon
  - Convince Elon Musk I can lead the mission to do this
  - Then successfully manage to establish the first base on the moon

  But how probable is this really?

Possibility vs Probability: Occam's razor

- Occam's razor: “the simplest explanation is usually the probable one”
  - NOTE: This is not a perfect rule, rather, it is a guiding heuristic

- Often, if you lack any evidence and are faced with two competing theories it may be best to first investigate the simplest
  - If for no other reason than it is simplest to first test
  - This does not mean you shouldn’t investigate the second

Casual Inference

What is causal inference: Understand what causes an outcome (e.g., Why is text entry poor on touchscreens?)

Correlation ≠ Causation

- Observing X and Y together doesn't mean X caused Y
- A hidden factor Z might cause both

Solution: Controlled Experiments

- Control other factors through design
- Compare the **treatment group** vs the **control group**
- Isolate the effect of your factor of interest

Casual Inference

Problem: Uncontrolled differences between groups (experience, physical traits, etc.)

Solutions (we cover this more tomorrow but):

1. **Randomised Control:** Random assignment, blind conditions
2. **Repeated Measures:** All participants experience all conditions

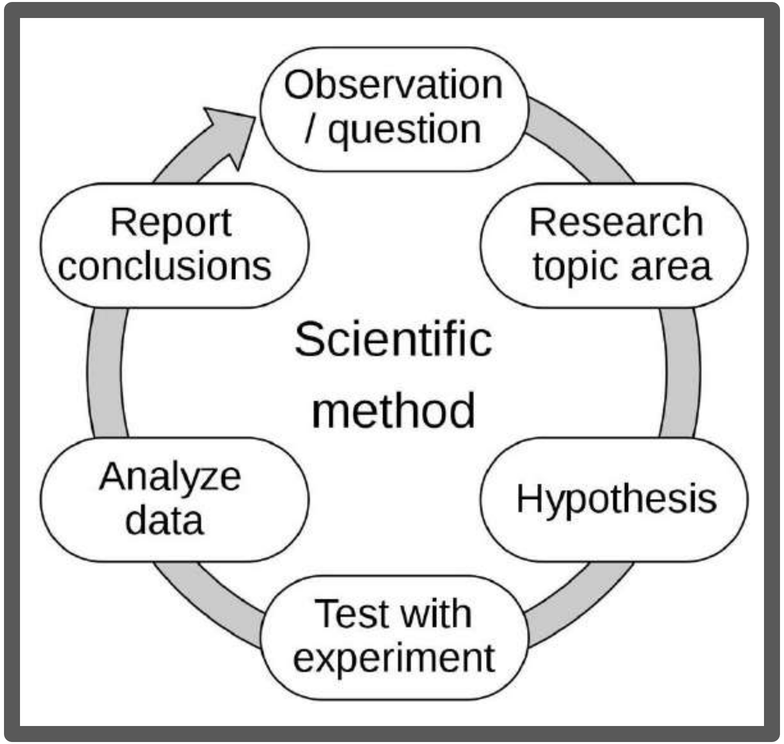
Bottom line: Proper experimental design allows safe causal inferences

Summary of Lecture 3 Part 1: Reading

- The Moodle videos focus on the scientific method (with a little philosophy)
- The importance of standing your ground but with the humility to be proved wrong
- That you can never know that you are completely right — only when you are completely wrong (and temporarily right)

Summary of Lecture 3 Part 1: Reading

- Euan Freeman’s video introduces the method in detail



Summary — The Scientific Method: Reading

1. Define a question
2. Gather information and resources (observe)
3. Form an explanatory hypothesis
4. Test the hypothesis by performing an experiment and collecting data in a reproducible manner
5. Analyze the data
6. Interpret the data and draw conclusions that serve as a starting point for a new hypothesis
7. Publish results
8. Retest (frequently done by other scientists)

Summary of Lecture 3 Part 1: Lecture

- We spoke about arguments (philosophically)
  - Inductive vs deductive inference
  - Informal and formal fallacies
  - False dilemmas
  - Straw man arguments

- Possibility vs probability

Any Questions?

Quick Break Before Part 2